

Alexander Belik

Theoretical Physics | Quantum Thermodynamics | Scientific Computing

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PROFILE

Theoretical physics graduate and current M.Sc. High-Performance Computing student at Trinity College Dublin. I build research platforms, GPU and HPC simulation tooling, machine-learning systems, and full-stack products across theoretical physics, scientific computing, and data-heavy software.

EDUCATION

M.Sc. High-Performance Computing

Trinity College Dublin, current

B.A. Theoretical Physics

Trinity College Dublin, 2022-2026

EXPERIENCE

Scientific Software Developer

Trinity College Dublin

May 2026 - Sep 2026

- Built a digital lab notebook platform storing experimental measurements as structured, auditable records.
- Developed GPU-accelerated physics demonstrations that run headlessly on the Leonardo EuroHPC system.

Teaching Assistant

Trinity College Dublin

Oct 2025 - May 2026

- Led weekly tutorials and problem-solving sessions aligned with lectures.
- Graded assignments and exams with clear, constructive feedback.

Research Assistant

Trinity College Dublin, CRANN Institute

Jun 2025 - Sep 2025

- Designed Echo State Network reservoir models for quantum time-evolution data.
- Analysed model stability, convergence, and predictive accuracy across hyperparameters.

Software Engineer (EdTech)

Grinds360

Jun 2024 - Dec 2025

- Built Python automation tools for video and interactive lesson workflows.
- Contributed to UX iteration in Figma and full-stack product features.

Experimental / Computational Research Assistant

Trinity College Dublin

May 2024 - Sep 2024

- Created a Python spectroscopy control system for ferrofluids and ferronematic liquid crystals.
- Built automated acquisition and analysis workflows for repeated experimental data.
- Contributed to liquid-crystal research that led to co-authored publication work.

Exam Corrector

The Dublin Academy of Education

Jul 2021 - Jan 2024

- Corrected exams and coursework and supported marking workflows.

Selected work

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CAPSTONE THESIS

Locally Thermal from Global Athermality

Quantum thermodynamics project on globally correlated states whose local subsystems are exactly thermal, with an SDP toolkit behind it.

- Characterized locally thermal states as affine slices of the positive semidefinite cone
- Derived a full nine-parameter two-qubit normal form

Tools: Quantum thermodynamics, Linear algebra, SDP methods, Python, LaTeX

PHYSICS-INFORMED ML

EchoState and Heisenberg Chain

PyTorch Echo State Network toolkit plus a Heisenberg spin-chain simulator for learning quantum time dynamics.

- General-purpose ESN package with configurable reservoirs, feedback, washout, and mixed precision
- Nine ridge-regression solvers plus streaming covariance accumulation

Tools: PyTorch, NumPy, SciPy, QuTiP, Optuna, Python

[View on GitHub](#)

HPC AND GPU SIMULATION

Leonardo Visual Demos

Thirteen GPU-accelerated physics demonstrations that run headlessly on the Leonardo supercomputer and replay through a browser viewer.

- Thirteen solvers spanning lattice-Boltzmann flow, particle-mesh cosmology, N-body galaxy collisions, reaction-diffusion, plasma control, and molecular dynamics
- One demo contract that runs on NumPy, CuPy/CUDA, or a hybrid pipeline overlapping GPU solving with CPU frame encoding

Tools: Python, CuPy / CUDA, PyTorch, NumPy, SLURM, JavaScript

[View on GitHub](#)

PUBLISHED EXPERIMENTAL RESEARCH

Ferronematic Liquid-Crystal Research

Python experiment control, spectroscopy data analysis, and ferronematic liquid-crystal research contributing to an RSC publication.

- Python control system driving a spectroscopy instrument through its Windows acquisition software
- Sequential sample imaging under stepped temperature and voltage control

Tools: Python, Instrumentation, Data acquisition, Matplotlib, Research workflow

[RSC publication](#)

COMPUTATIONAL CHEMISTRY

DFT Hamilton

Containerised PySCF environment for density-functional and Hartree-Fock experiments, with functional and spin-correlation comparisons.

- Docker environment with one-command setup, verifying basis loading, RHF, LibXC DFT, and open-shell UHF
- Scripted molecular example runs that write CSV and Markdown result tables

Tools: Python, PySCF, LibXC, Docker, NumPy